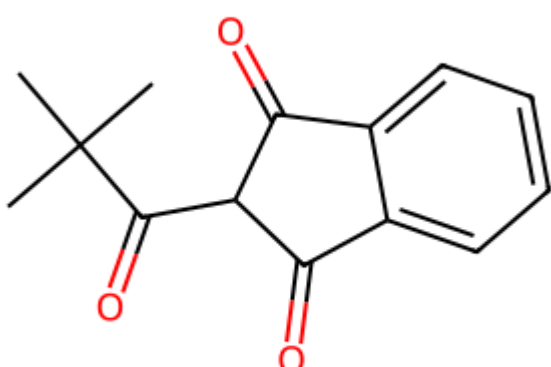


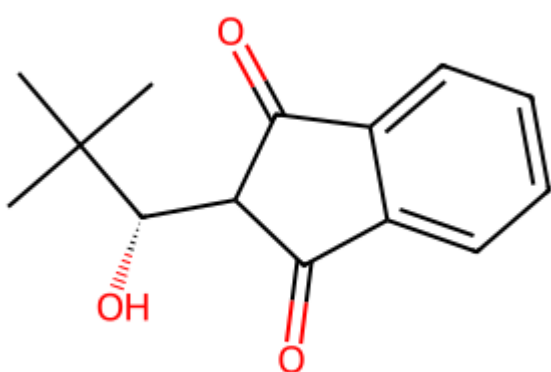


```
{'analysis_time': 0.0,  
'first_solution_time': 0.641751,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 2.8264959563086918,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 2.8264959563086918,  
'retro_star_value': 2.8264959563086918}
```



Reaction

```
{'analysis_time': 0.118696,  
'first_solution_time': 2.749889,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 6.2823517505087105,  
'retro_star_reaction_number': 6.2823517505087105,  
'retro_star_rxn_cost': 5.809142990314027,  
'retro_star_value': 6.2823517505087105}
```

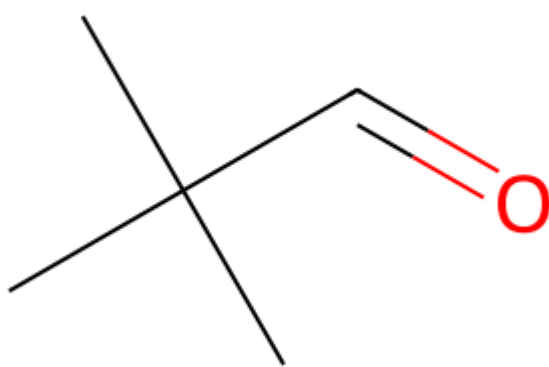
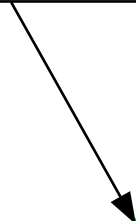
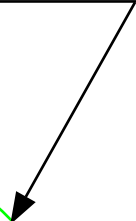


```
{'analysis_time': 0.118696,  
'first_solution_time': 2.749889,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 0.4732087601946839,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 0.4732087601946839,  
'retro_star_value': 6.2823517505087105}
```

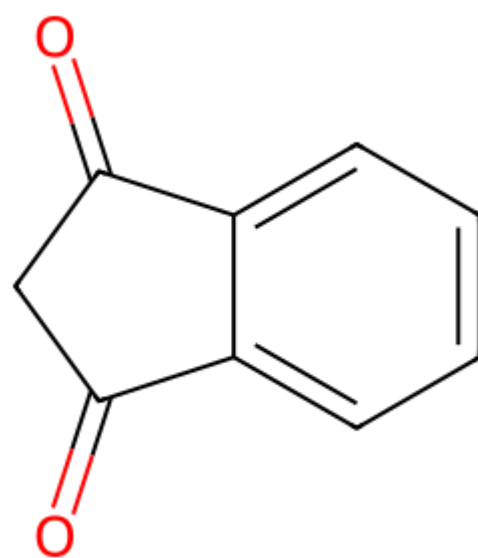


Reaction

```
{'analysis_time': 2.749889,  
'first_solution_time': 2.749889,  
'num_calls_rxn_model': 27,  
'num_calls_value_function': 726,  
'retro_star_min_cost': 0.4732087601946839,  
'retro_star_reaction_number': 0.4732087601946839,  
'retro_star_rxn_cost': 0.4732087601946839,  
'retro_star_value': 6.2823517505087105}
```



```
{'analysis_time': 2.749889,  
'first_solution_time': 2.749889,  
'num_calls_rxn_model': 27,  
'num_calls_value_function': 726,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 6.2823517505087105}
```



```
{'analysis_time': 2.749889,  
'first_solution_time': 2.749889,  
'num_calls_rxn_model': 27,  
'num_calls_value_function': 726,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 6.2823517505087105}
```